

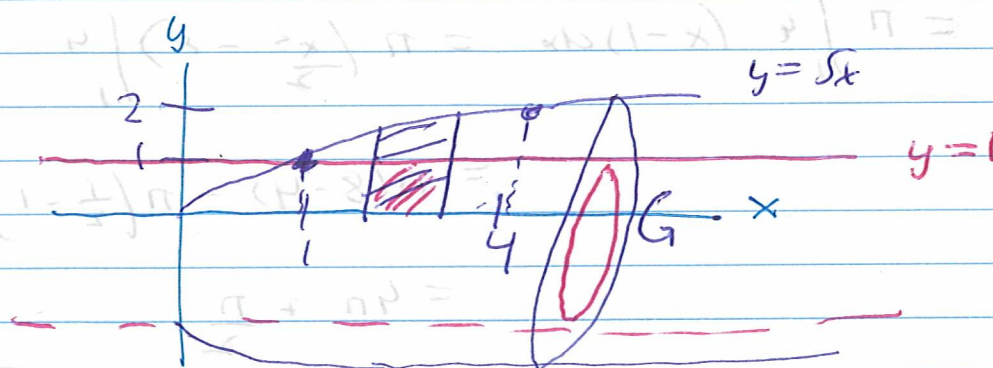
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Lesson 21 Solids of Revolution: Washer Method

We continue computing volumes of solids of revolution.

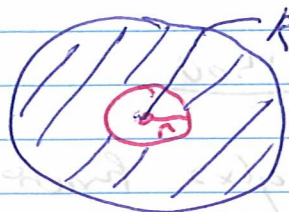
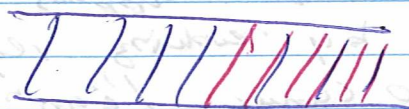
Let's consider an example with a revolving region between two curves.

Ex1 What is the volume of the solid formed by revolving region between $y = \sqrt{x}$ (blue) and $y = 1$ over $[1, 4]$ about the x-axis? (pink)



x	y = $\sqrt{x}$
0	0
1	1
4	2

The cross section here changes from a disk to a washer



Washer

$$\text{Area} = \pi R^2 - \pi r^2$$



Thickened washer

$$\text{Volume} = (\pi R^2 - \pi r^2) dx$$

$$R = \sqrt{x} - 0 = \sqrt{x}$$

$$r = 1 - 0 = 1$$

$\therefore$

$$\text{Volume} = \int_1^4 \pi (\sqrt{x})^2 - 1^2) dx$$

$$= \pi \int_1^4 (x-1) dx = \pi \left(\frac{x^2}{2} - x \right) \Big|_1^4$$

$$= \pi(8-4) - \pi\left(\frac{1}{2} - 1\right)$$

$$= 4\pi + \frac{\pi}{2}$$

$$= \frac{8\pi + \pi}{2} = \boxed{\frac{9\pi}{2}}$$

* Washer method applies to solids formed by revolving region between two curves / "gap" between region & axis of rotation.

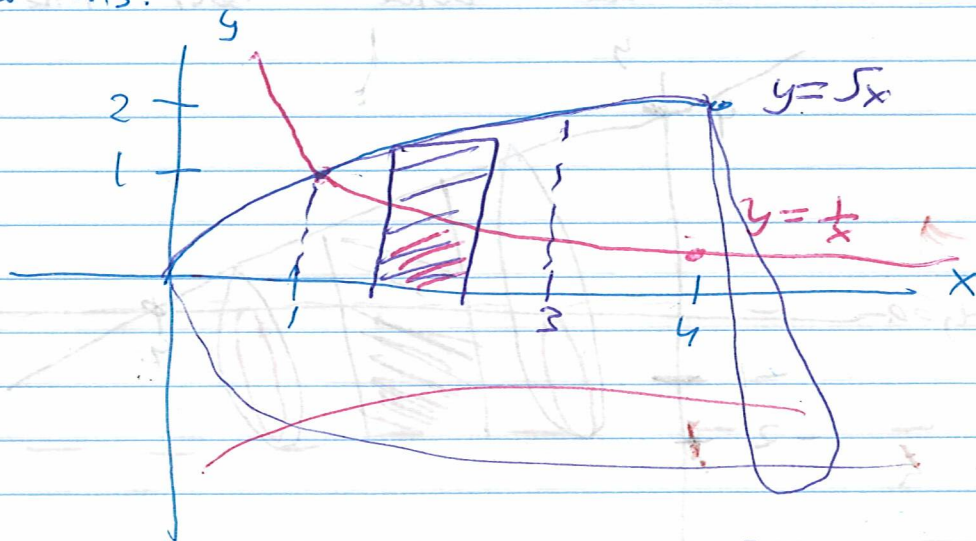
Washer Method

If $f(x)$, $g(x)$ functions, with $f(x) \geq g(x)$ for x in $[a, b]$, then volume of solid formed by revolving region between $f(x)$ & $g(x)$ over $[a, b]$ about

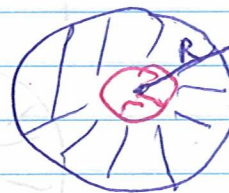
x -axis is

$$\int_a^b \pi ((f(x))^2 - (g(x))^2) dx$$

Ex. 2 Find volume of solid formed by revolving region bounded by $\sqrt{x}$ and $\frac{1}{x}$ over interval $[1, 3]$ about x -axis.



x	$1/x$
1	1
4	$1/4$



Washer

$$r(x) = \frac{1}{x} - 0 = \frac{1}{x}$$

$$R(x) = \sqrt{x} - 0 = \sqrt{x}$$

$$\text{Volume} = \int_1^3 \pi \left((\sqrt{x})^2 - \left(\frac{1}{x}\right)^2 \right) dx$$

$$= \pi \int_1^3 \left(x - \frac{1}{x^2} \right) dx$$

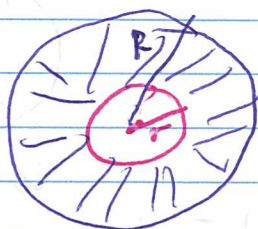
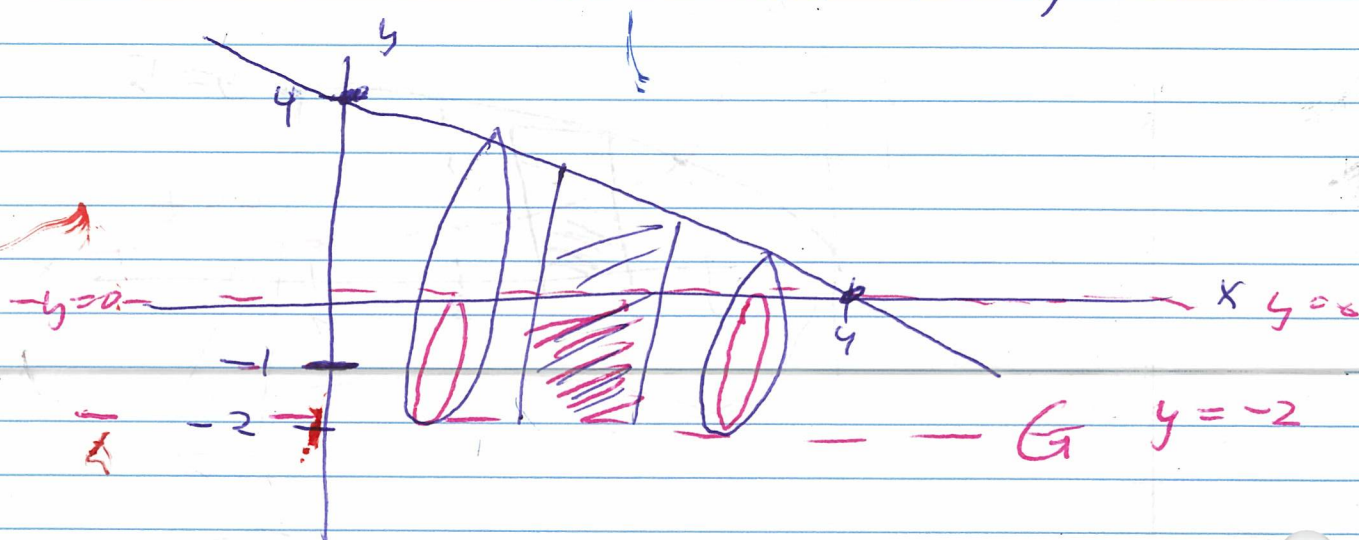
$$= \pi \left(\frac{x^2}{2} + x^{-1} \right) \Big|_1^3$$

$$= \pi \left(\frac{9}{2} + \frac{1}{3} - \frac{1}{2} - 1 \right)$$

$$= \pi \left(4 - 1 + \frac{1}{3} \right) = \pi \left(3 + \frac{1}{3} \right) = \left[\frac{10}{3} \pi \right]$$

Ex. 3. Find the volume of solid formed by revolving region bounded by $4-x$ and x -axis about the line $y=-2$ over $[0,4]$

this time we revolve about line $y=-2$.



Washer since the line is $y = -2$ for rotation

$$R(x) = (4-x) - (-2) = 4-x+2 = 6-x$$

$$r(x) = 0 - (-2) = 2$$

$$\text{Volume} = \int_0^4 \pi ((6-x)^2 - 2^2) dx$$

$$= \pi \int_0^4 (36 - 12x + x^2 - 4) dx$$

$$= \pi \int_0^4 (32 - 12x + x^2) dx$$

$$= \pi \left[32x - \frac{12x^2}{2} + \frac{x^3}{3} \right] \Big|_0^4$$

$$= \pi \left[32x - 6x^2 + \frac{x^3}{3} \right] \Big|_0^4$$

$$= \pi \left[32(4) - 6(4)^2 + \frac{4^3}{3} - 0 \right]$$

$$= \pi \left[128 - 96 + \frac{64}{3} \right]$$

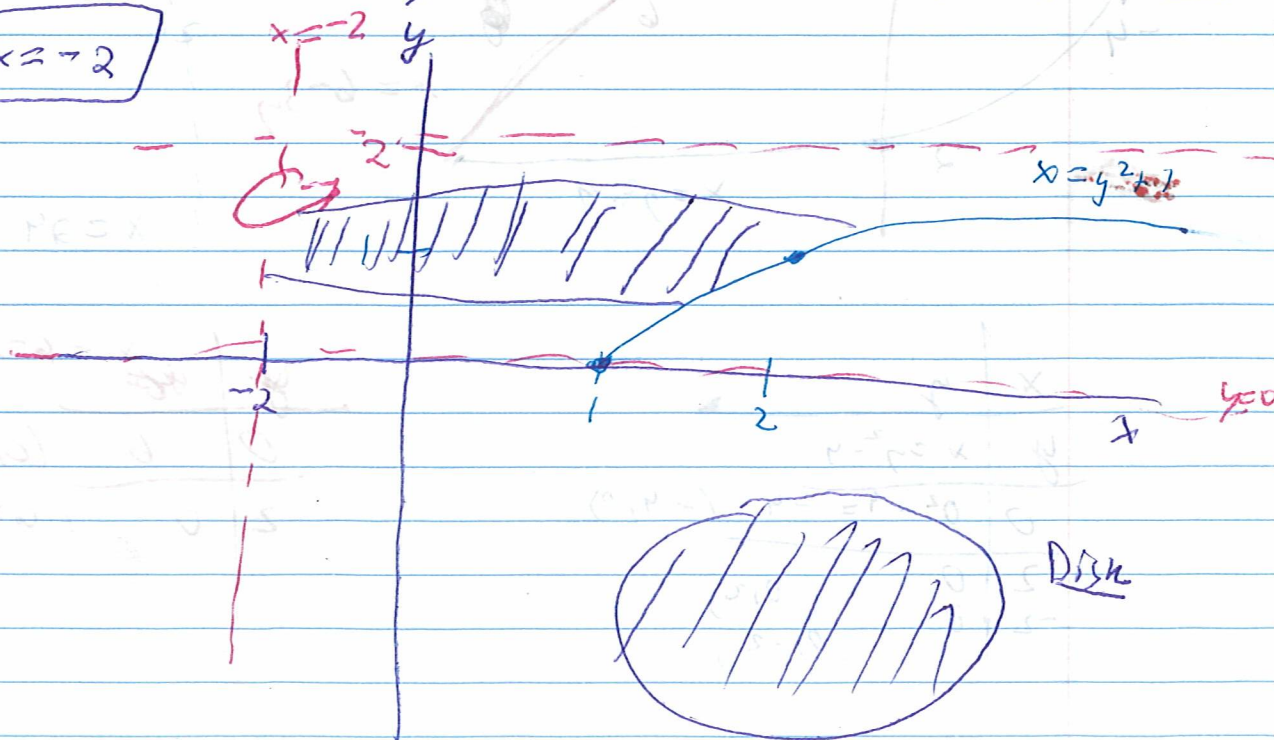
$$= \pi \left[32 + \frac{64}{3} \right]$$

$$= \boxed{\frac{160\pi}{3}}$$

Ex. 4 Find the volume of the solid that results when region enclosed by $x = y^2 + 1$, $x = -3$, $y = 0$, $y = 2$ about

$$\boxed{x = -2}$$

y	$x = y^2 + 1$
0	1 (1, 0)
1	2 (2, 1)
-1	2



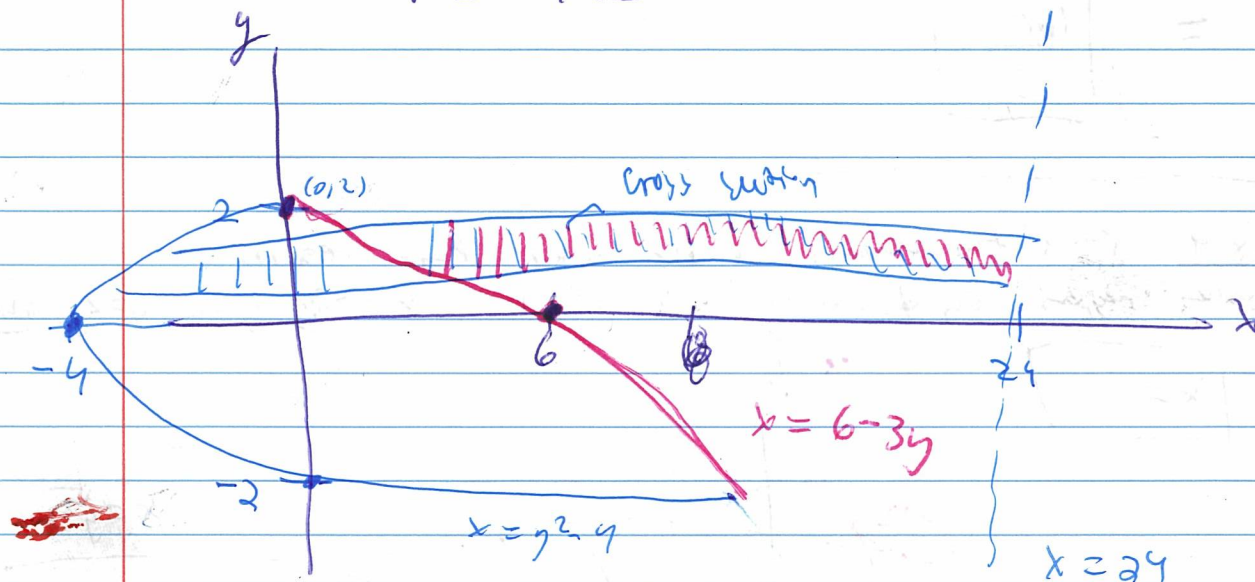
$$R(y) = (y^2 + 1) - (-2) = y^2 + 3$$

Disk method: $\text{Volume} = \pi \int_0^2 (y^2 + 3)^2 dy$

$$\pi \int_0^2 (y^4 + 6y^2 + 9) dy = \boxed{\frac{202}{5} \pi} \approx 126.92$$

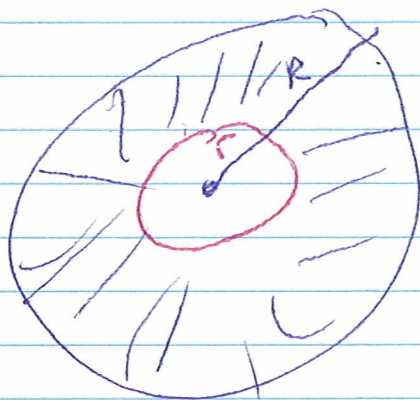
Ex. 8: Determine volume of the solid obtained by rotating region bounded by

(Blue) $x = y^2 - 4$ and (Red) $x = 6 - 3y$ about the line $x = 24$.



x	y
y	$x = y^2 - 4$
0	$0^2 - 4 = -4$ $(-4, 0)$
2	0 $(0, 2)$
-2	0 $(0, -2)$

x	y
0	6 $(6, 0)$
2	0 $(0, 2)$



Washer

outer
radius

$$R(y) = 24 - (y^2 - 4) = 28 - y^2$$

inner
radius

$$r(y) = 24 - (6 - 3y) = 18 + 3y$$

intersection points: $y^2 - 4 = 6 - 3y$

$$\Leftrightarrow y^2 + 3y - 10 = 0$$

$$(y + 5)(y - 2) = 0$$

$$\Rightarrow y = -5, y = 2$$

$$\text{Volume} = \pi \int_{-5}^2 \left[(28 - y^2)^2 - (18 + 3y)^2 \right] dy$$

$$= \pi \int_{-5}^2 (460 - 108y - 65y^2 + y^4) dy$$

$$= \pi \left(460y - 54y^2 - \frac{65}{3}y^3 + \frac{1}{5}y^5 \right) \Big|_{-5}^2$$

$$= \frac{31556}{15} \pi$$

